PC3SF11YVZA/ PC3SF11YVZB

Reinforced Insulation Type Phototriac Coupler for Triggering

■ Features

1. Isolation voltage between input and output ($V_{iso\ (rms)}$:5kV)
2. High critical rate of rise of OFF-state voltage (dV/dt:MIN. 1 000V/μs)
3. Internal isolation distance (0.4mm or more)
4. Recognized by UL (File No. E64380)
   Approved by VDE (VDE0884, File No.127413)
   Approved by BSI (BS415, File No.6690, BS7002, File No.7421)
   Approved by SEMKO (File No.0033029/01-04)
   Approved by DEMKO (File No.310107-01)
   Approved by FIMKO (File No.15795)

❊ **PC3SF11YVZA**, **PC3SF11YVZB** are for 200V line

■ Applications

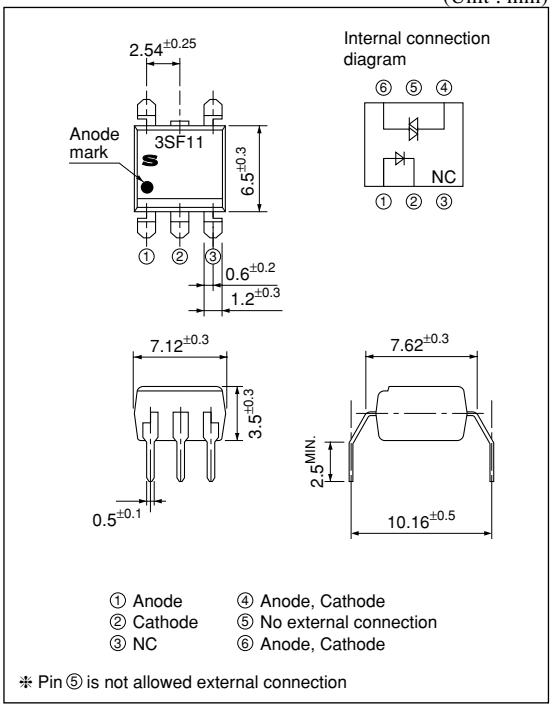
1. Home appliances
2. OA equipment, FA equipment
3. SSRs

■ Model Line-up

Minimum trigger current ($I_{FT[MAX.]}$)	for AC 200V line
10mA	**PC3SF11YVZA**
7mA	**PC3SF11YVZB**

■ Outline Dimensions

(Unit : mm)



❊ Pin ⑤ is not allowed external connection

■ Absolute Maximum Ratings

(Ta=25°C)

Parameter		Symbol	Rating	Unit
Input	*1 Forward current	I_F	50	mA
	Reverse voltage	V_R	6	V
Output	*1 RMS ON-state current	$I_{T\ (rms)}$	0.1	A
	Peak one cycle surge current	I_{surge}	1.2 (50Hz sine wave)	A
	Repetitive peak OFF-state voltage	V_{DRM}	600	V
*2 Isolation voltage		$V_{iso\ (rms)}$	5	kV
Operating temperature		T_{opr}	−30 to +100	°C
Storage temperature		T_{stg}	−55 to +125	°C
Soldering temperature		T_{sol}	260 (For 10s)	°C

*1 The derating factors of absolute maximum ratings due to ambient temperature are shown in Fig.1, 2

*2 AC for 1 min, 40 to 60%RH, f=60Hz

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Internet Internet address for Electronic Components Group http://sharp-world.com/ecg/

■ Electro-optical Characteristics

(Ta=25°C)

Parameter			Symbol	Conditions	MIN.	TYP.	MAX.	Unit
Input	Forward voltage		V_F	I_F=20mA	–	1.2	1.4	V
	Reverse current		I_R	V_R=3V	–	–	10^{-5}	A
Output	Repetitive peak OFF-state current		I_{DRM}	V_D=V_{DRM}	–	–	10^{-6}	A
	ON-state voltage		V_T	I_T=0.1A	–	–	2.5	V
	Holding current		I_H	V_D=6V	0.1	–	3.5	mA
	Critical rate of rise of OFF-state voltage		dV/dt	V_D=1/$\sqrt{2}$ • V_{DRM}	1 000	2 000	–	V/µs
Transfer characteristics	Minimum trigger current	**PC3SF11YVZA**	I_{FT}	V_D=6V, R_L=100Ω	–	–	10	mA
		PC3SF11YVZB			–	–	7	
	Isolation resistance		R_{ISO}	DC=500V, 40 to 60%RH	5×10^{10}	10^{11}	–	Ω
	Turn-on time		t_{on}	V_D=6V, R_L=100Ω, I_F=20mA	–	–	100	µs

Fig.1 RMS ON-state Current vs. Ambient Temperature

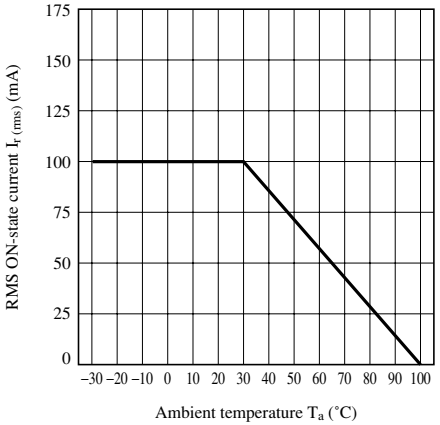


Fig.2 Forward Current vs. Ambient Temperature

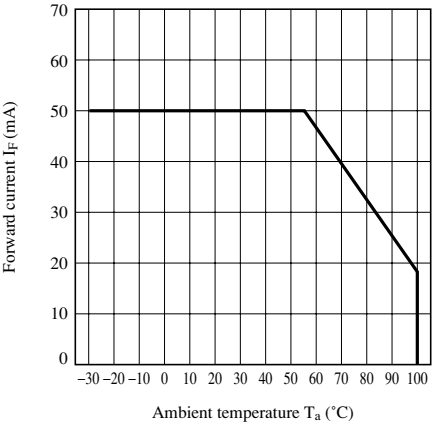


Fig.3 Forward Current vs. Forward Voltage

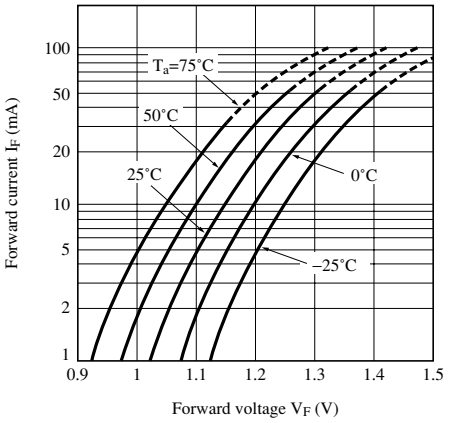


Fig.4 Minimum Trigger Current vs. Ambient Temperature

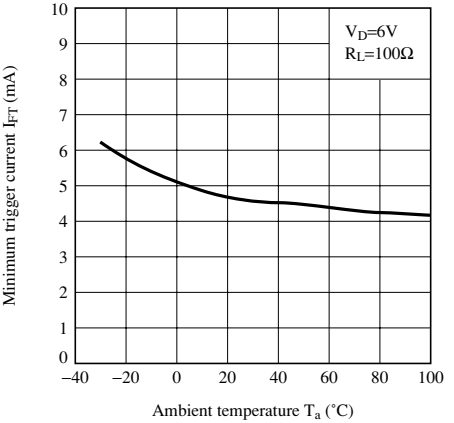


Fig.5 ON-state Voltage vs. Ambient Temperature

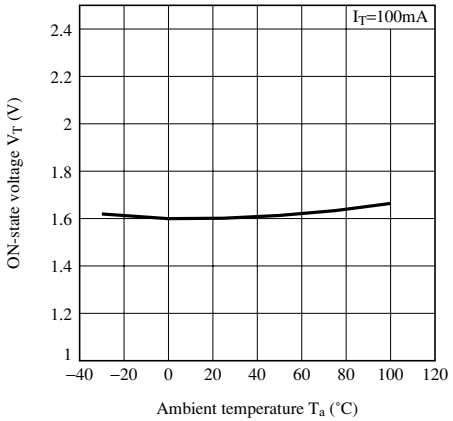


Fig.6 Holding Current vs. Ambient Temperature

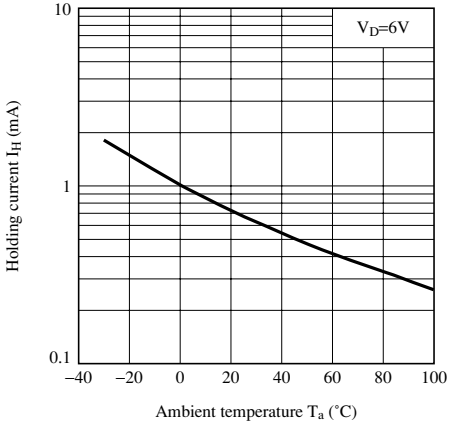


Fig.7 Repetitive Peak OFF-state Current vs. Ambient Temperature

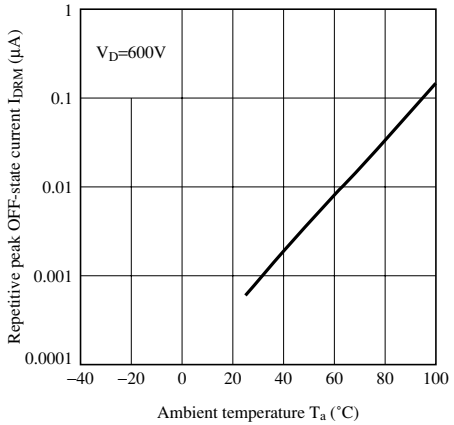


Fig.8 Relative Repetitive Peak OFF-state Voltage vs. Ambient Temperature

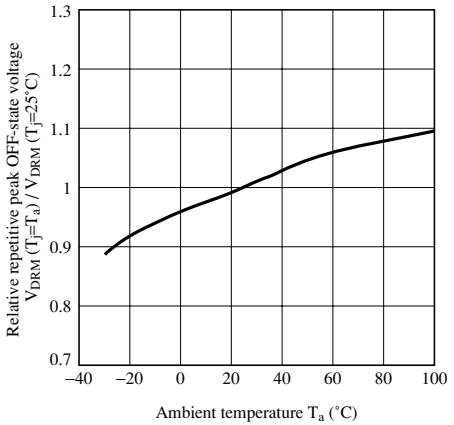
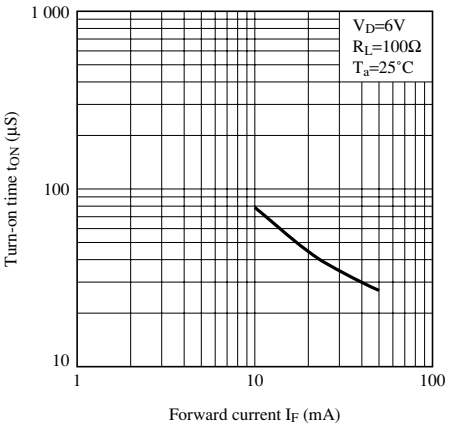


Fig.9 Turn-on Time vs. Forward Current



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  --- Office automation equipment
  --- Telecommunication equipment [terminal]
  --- Test and measurement equipment
  --- Industrial control
  --- Audio visual equipment
  --- Consumer electronics

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  --- Gas leakage sensor breakers
  --- Alarm equipment
  --- Various safety devices, etc.

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